Institute of Information Technology
University of Dhaka

Topic: Goal Question Metrics (GQM)
Software Metrics (SE 611)

Investigating the societal and personal factors contributing to the deterioration of mental health among IIT DU students

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1. Introduction

GQM is the abbreviation for "goal, question, metric". It is an established goal-oriented approach to software metrics to improve and measure software quality.

GQM defines a measurement on three dimensions. They are-

Conceptual Level (Goal): A goal is defined for an object, for a variety of reasons,with respect to various models of quality, from various points of view and relative to a particular environment.

Operational level (Question): A set of questions is used to define models of the object of study and then focuses on that object to characterize the assessment or Achievement of a specific goal.

Quantitative level (Metric): A set of metrics, based on the models, is associated With every question in order to answer it in a measurable way.

The first step is deciding our goal, then develop a question set to collect data from stakeholders. The questions must have appropriate metrics to measure the result later. We have to create a questionnaire to collect data from people defined in the scope. The last part will include metrics analysis with result derivation to finally reach an outcome.

2. Project Specifications

2.1 Overview

Mental health is a critical factor in a student's academic performance and overall well-being. Software measurement, especially with the Goal-Question-Metric (GQM) approach, helps systematically assess such intangible aspects.

2.2 Motivation

The GQM approach allows for a structured investigation into mental health issues by defining clear goals, deriving questions, and establishing measurable metrics. This ensures that the subjective and sensitive nature of mental health is explored rigorously.

2.3 Scope

This assignment focuses on identifying and evaluating the societal and personal factors that impact the mental health of IIT DU students. Through survey responses, key trends and patterns will be analyzed to support institutional improvements.

3. Goal Specification

3.1 GQM Framework

The GQM approach consists of three levels: Goal (what you want to achieve), Question (what you need to ask to know if you're achieving the goal), and Metric (what data is used to answer the questions). It is particularly useful in contexts like this where qualitative and quantitative factors interact.

General Statement: Investigating the societal and personal factors contributing to the deterioration of mental health among IIT DU students by measuring the frequency of academic stress, lifestyle patterns, institutional support, and student-identified stressors.

3.2 PPE Approach

Purpose: To evaluate how personal responsibilities, academic challenges, and institutional limitations contribute to the mental health condition of IIT DU students.

Perspective: To understand the lived experiences of students from different academic years, gender identities, and socio-economic backgrounds in relation to mental well-being.

Environment: This study focuses on students currently enrolled in the Institute of Information Technology (IIT), University of Dhaka, who reside in diverse environments including on-campus, off-campus, and at home, and who are navigating varying academic and social pressures.

After specifying our goal through the purpose-perspective-environment framework, our final goal is -

“To explore and evaluate the personal and societal stressors that contribute to poor mental health among IIT DU students, considering academic, lifestyle, and institutional factors, from the students’ own perspectives and environments.”

3.3 Subgoals

1. Assess the impact of academic pressure on students' mental health.
2. Evaluate the role of personal responsibilities and lifestyle factors.
3. Explore students’ perception of mental health awareness and support systems.

4. Identify the top self-reported factors negatively affecting mental health.

3.4 Questions and Metrics

Subgoal 1: Assess the impact of academic pressure on students' mental health

Q1. How often do students feel overwhelmed by their academic workload?

- **M1.** Frequency scale response (Never, Sometimes, Often, Always) – *Qualitative*
- **M2.** Textual elaboration from free-response fields – *Qualitative*

Q2. How many hours do students spend on academic work daily?

- **M1.** Multiple-choice (e.g., "<3 hours", "3–5 hours", ">5 hours") – *Quantitative*

Q3. How clearly are academic deadlines and expectations communicated?

- **M1.** Subjective response on clarity scale – *Qualitative*

Subgoal 2: Evaluate the role of personal responsibilities and lifestyle factors

Q1. Do students have part-time or full-time jobs alongside their studies?

- **M1.** Binary response (Yes/No) – *Quantitative*

Q2. How far is the student's residence from their institution?

- **M1.** Categorical data (Less than 1 km, 1–5 km, etc.) – *Quantitative*

Q3. How much sleep do students typically get during academic terms?

- **M1.** Self-reported hours per night – *Quantitative*
- **M2.** Correlation with stress level – *Derived/Quantitative*

Q4. How does commuting or financial burden affect students' stress levels?

- **M1.** Descriptive responses – *Qualitative*
- **M2.** Occurrence of financial-related issues in open-ended questions – *Qualitative*

Subgoal 3: Explore students' perception of mental health awareness and support

Q1. Do students think mental health should be part of the curriculum?

- **M1.** Yes/No – *Quantitative*
- **M2.** Justification provided – *Qualitative*

Q2. Are students willing to attend mental health workshops?

- **M1.** Response frequency (Always, Sometimes, Never) – *Qualitative*

Q3. Do students believe peer-led initiatives could reduce stigma?

- **M1.** Categorical opinion (Yes, No, Not Sure) – *Qualitative*

Q4. Have students used mental health apps or consulted professionals?

- **M1.** Binary responses (Yes/No for each) – *Quantitative*
- **M2.** Follow-up suggestions in open responses – *Qualitative*

Subgoal 4: Identify the top self-reported factors negatively affecting mental health

Q1. What societal factors (e.g., family expectations, social pressure) are commonly reported?

- **M1.** Keyword frequency in open responses – *Qualitative*

Q2. What personal challenges (e.g., procrastination, time management) appear repeatedly?

- **M1.** Recurring phrases in text analysis – *Qualitative*

Q3. What role does isolation or lack of peer connection play?

- **M1.** Mentions of loneliness or lack of support – *Qualitative*

Q4. What institutional improvements do students recommend?

- **M1.** Categorized suggestions (e.g., Counseling, Deadline policies) – *Qualitative*

- **M2.** Sentiment tone of open-ended suggestions – *Qualitative*

4. Questionnaire Preparation and Data Collection

4.1 Data Collection Methods

The data was collected through a Google Forms survey distributed among students of IIT DU. The form included both multiple-choice and open-ended questions targeting academic, personal, and societal factors influencing mental health. Participants provided responses anonymously to ensure honesty and confidentiality. A total of **35** responses were gathered during the month of January 2025. Proof of data collection includes timestamped form submissions and access to the raw spreadsheet file.

4.2 Tools Used

The primary tools used were:

- **Google Forms** – for creating and distributing the questionnaire.
- **Google Sheets / Excel** – for collecting and organizing the response data.
- **Manual review** – for extracting insights from open-ended responses.

*No automated tools or scripts were used in this version of the report for data visualization or analysis.

Some sample questions based on the subgoals are given below-

Subgoal 1: Assess the impact of academic pressure on students' mental health

1. How do you feel when you receive negative academic feedback (e.g., low grades)?

- Motivated to do better
- Anxious or stressed
- Depressed or disheartened
- Indifferent

2. How many hours do you spend on academic work daily?

- Less than 3 hours
- 3–5 hours
- More than 5 hours

3. How often do you feel overwhelmed by your academic workload?

- Never
- Sometimes
- Often
- Always

4. Have you ever considered dropping out of a course or program due to academic pressure?

- Yes
- No
- Maybe
-

Subgoal 2: Evaluate the role of personal responsibilities and lifestyle factors

1. How far is your residence from your institution?

- Less than 1 km
- 1-5 km
- More than 5km

2. Do you have any part-time or full-time job alongside your studies?

- Yes
- No

3. Do you feel pressured to achieve higher grades due to family expectations?

- Yes
- No
- Sometimes

4. How often do you feel homesick or miss your family?

- Never
- Occasionally
- Often
- Always

Subgoal 3: Explore students' perception of mental health awareness and support

1. If given the option, would you attend mental health workshops regularly?

- Always
- Sometimes
- Not at all

2. Do you believe peer-led mental health initiatives would help reduce mental health stigma?

- Yes
- No
- Not sure

3. Do you think mental health should be included as part of the curriculum?

- Yes
- No

4. How do you compare yourself to your peers academically and socially?

- Never
- Rarely
- Sometimes
- Often
- Always

Subgoal 4: Identify the top self-reported factors negatively affecting mental health

1.What contributes the most to your mental health struggle?

- Academic pressure
- Family expectations
- Social expectations
- Personal expectations

2.Do you often compare yourself to people you see on social media?

- Yes
- No
- Sometimes

3.Do you feel that using technology late at night disrupts your sleep schedule?

- Yes
- No
- Sometimes

4.**What are the top three factors that worsen students' mental health?**

-Description

Full questionnaire can be found here:
https://docs.google.com/forms/d/1G03u588hNIfrPolIsPbTB_Jf6hwI0nWPj-gC_x03W2A/edit#responses

5. Data Visualization

The summary of the collected data is presented here in graphs and charts:

Subgoal 1: Assess the impact of academic pressure on students' mental health

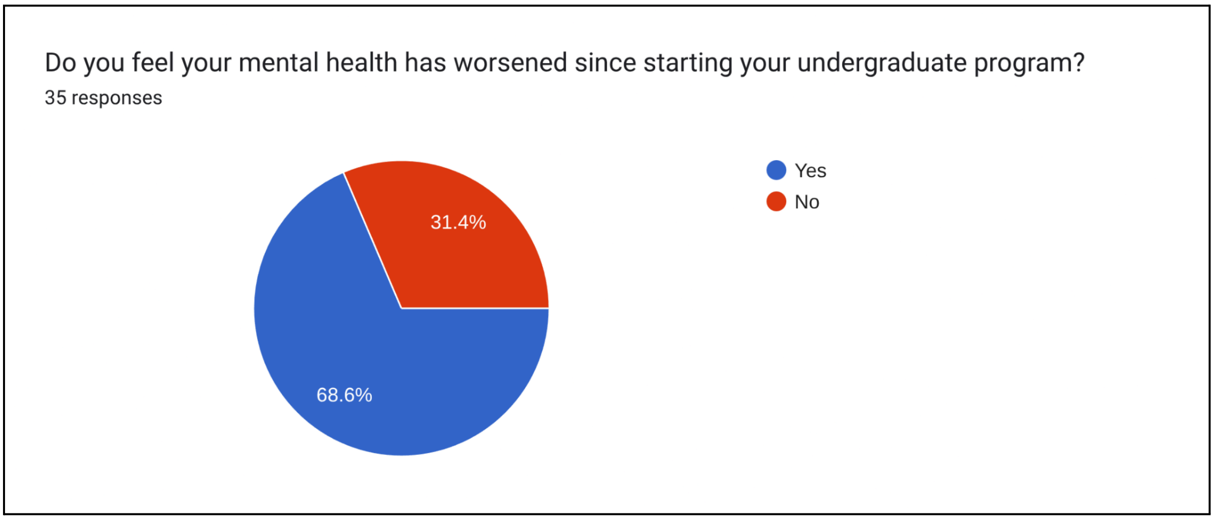


Figure 1-Respondent's mental health situation after starting undergrad.

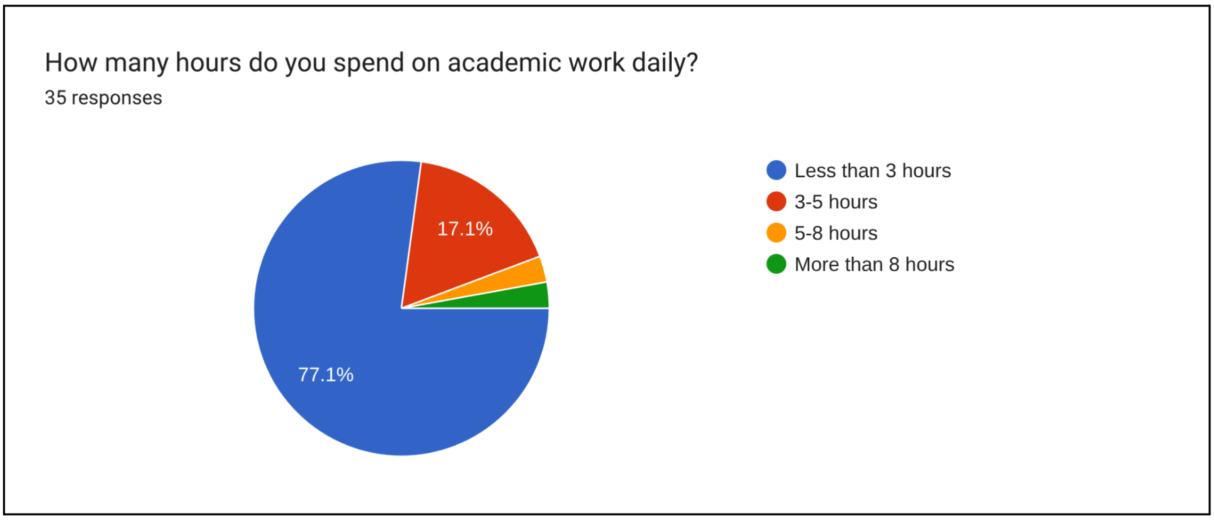


Figure 2.- Respondent's hours daily on academic work.

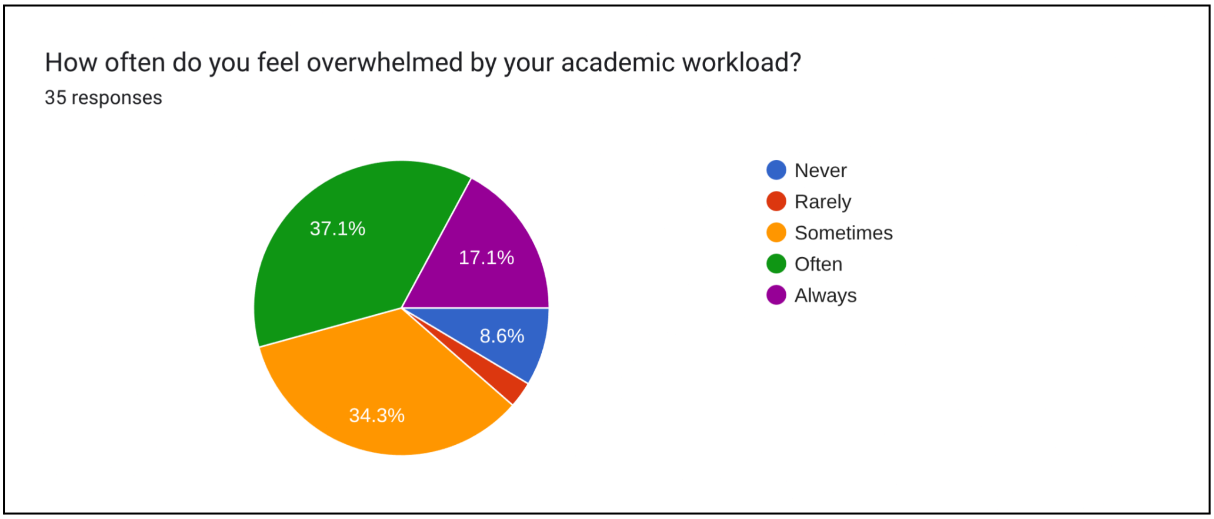


Figure 3. Pressure of academic workload on respondents.

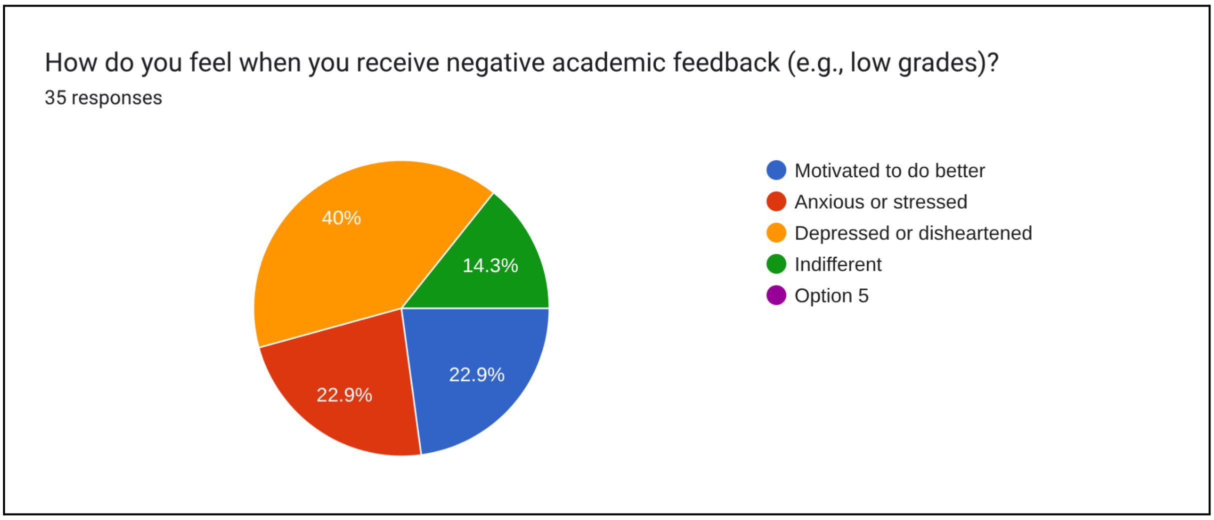


Figure 4. Impact of negative academic feedback on respondents.

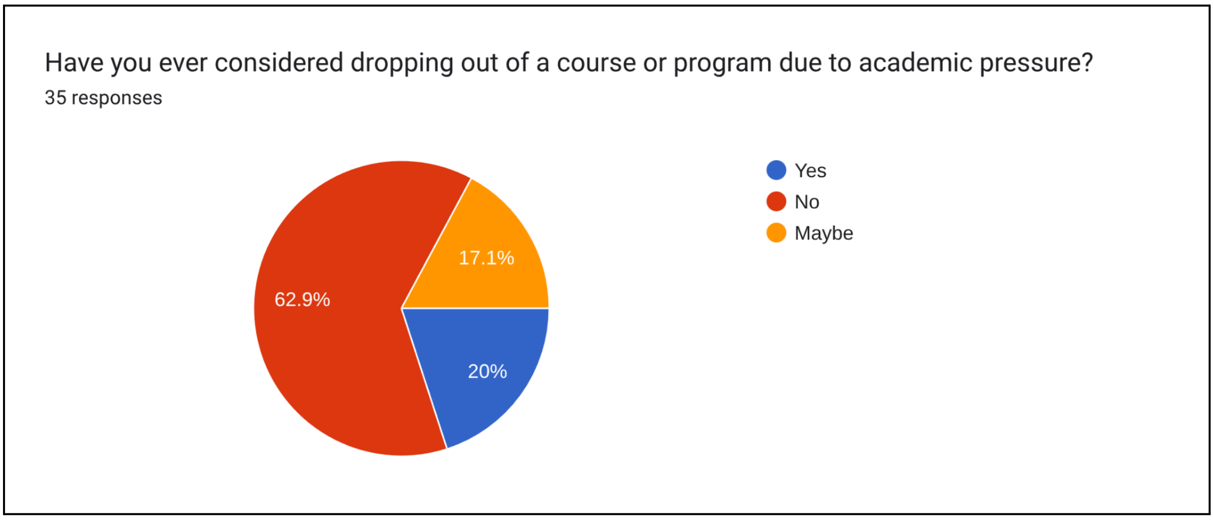


Figure 5. Respondents' likelihood of dropout due to academic pressure.

Subgoal 2: Evaluate the role of personal responsibilities and lifestyle factors

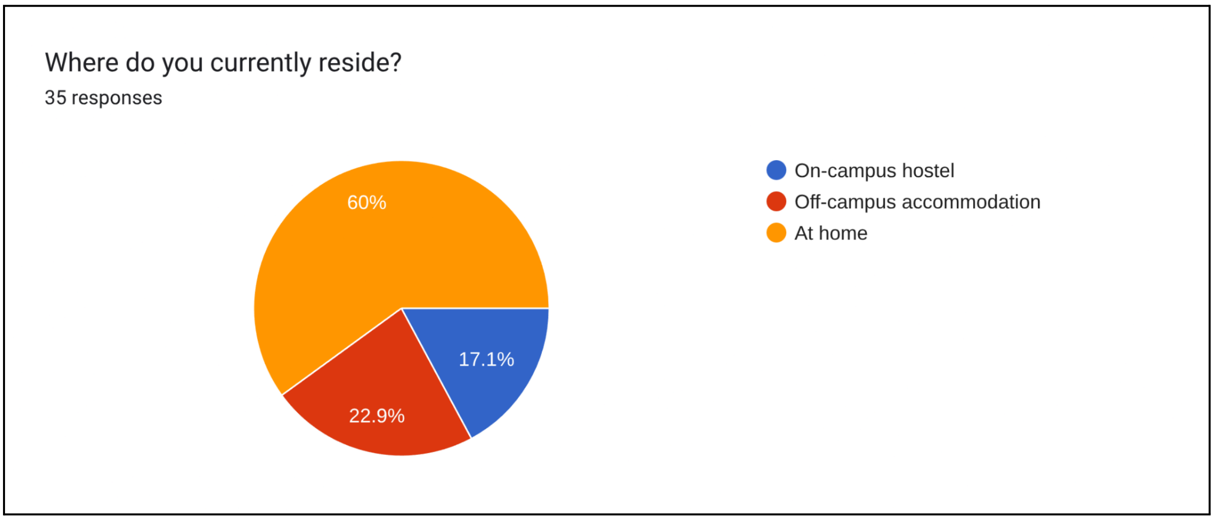


Figure 6. Respondents' residence type.

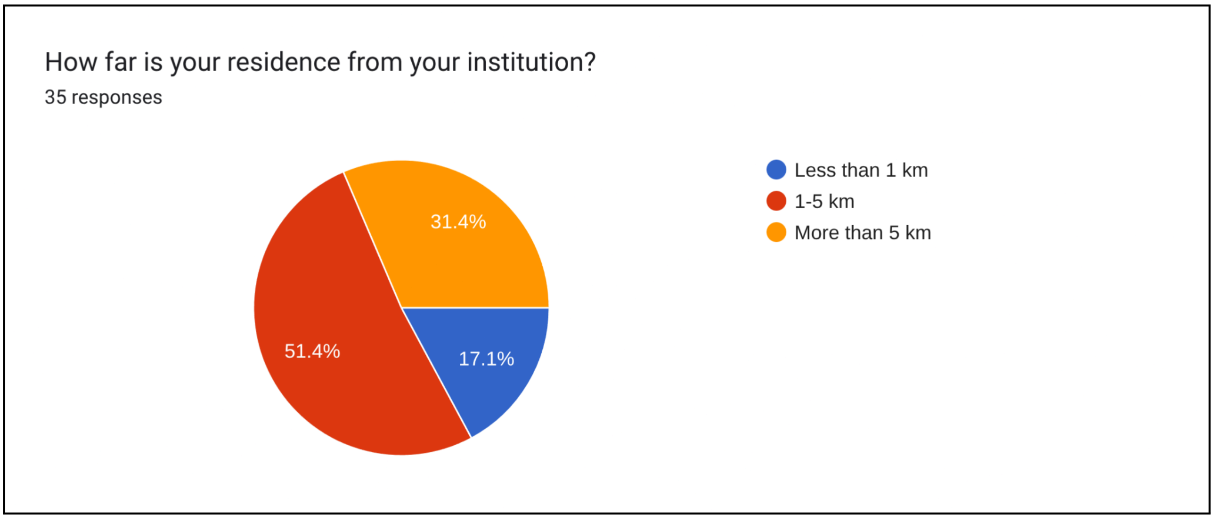


Figure 7. Distance of respondents' residence from institution.

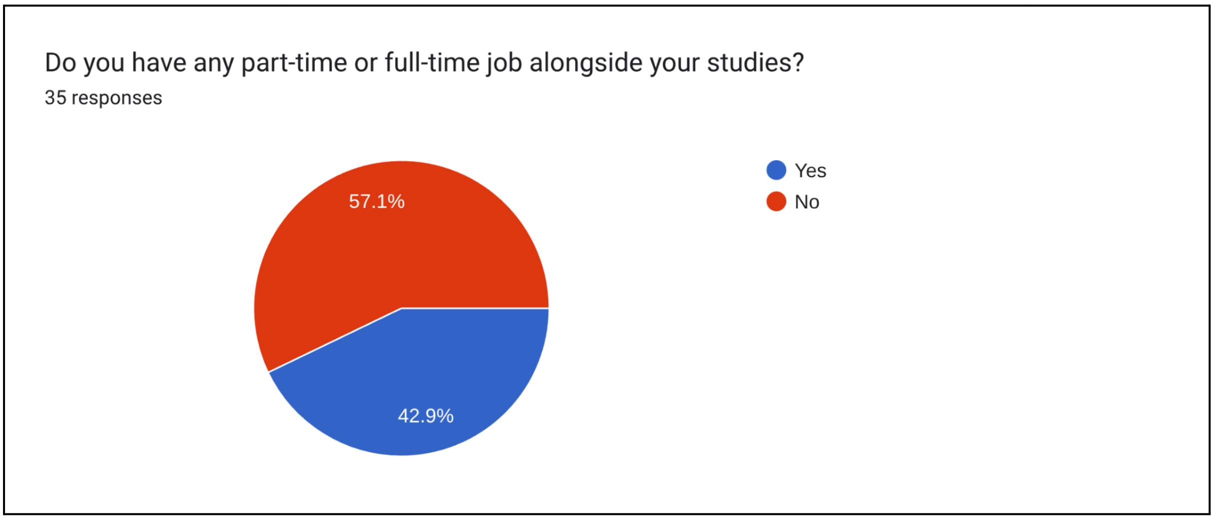


Figure 8. If respondents have a job alongside study.

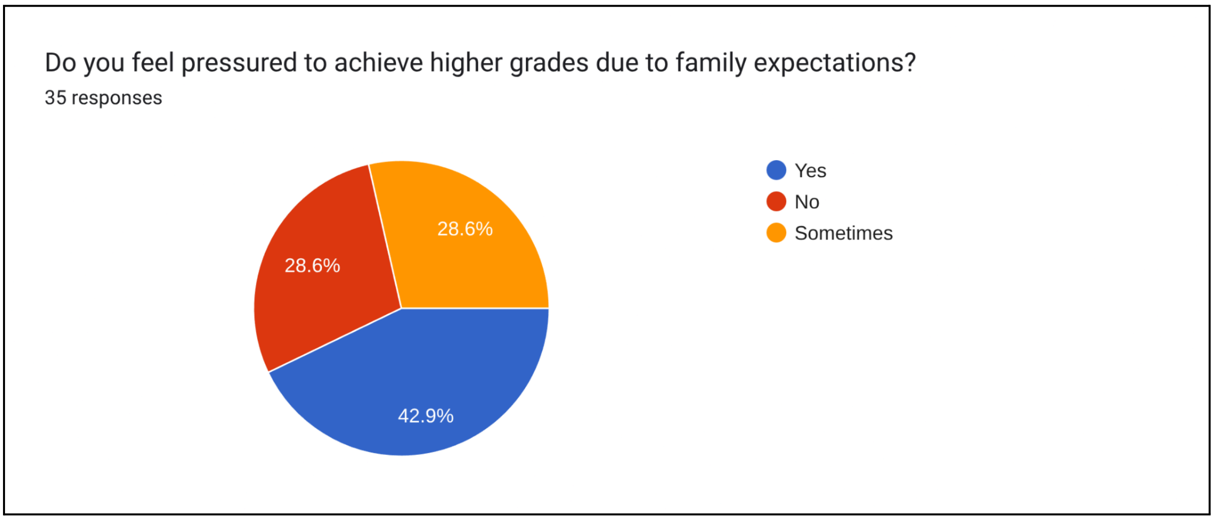


Figure 9. Pressure on respondents to achieve higher grades.

Subgoal 3: Explore students' perception of mental health awareness and support

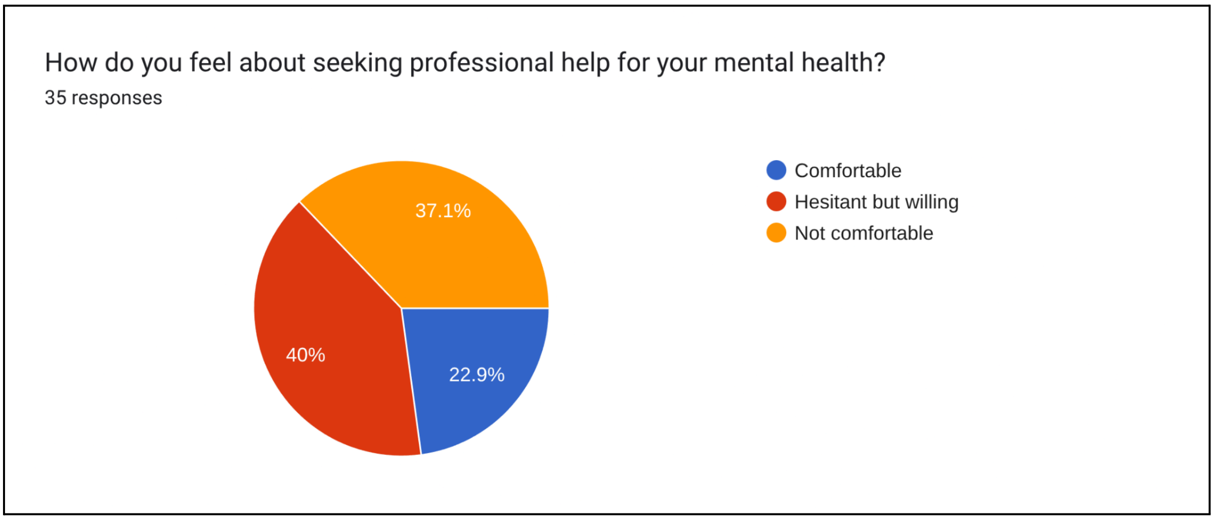


Figure 10. Respondent's perspective on seeking professional help on mental health.

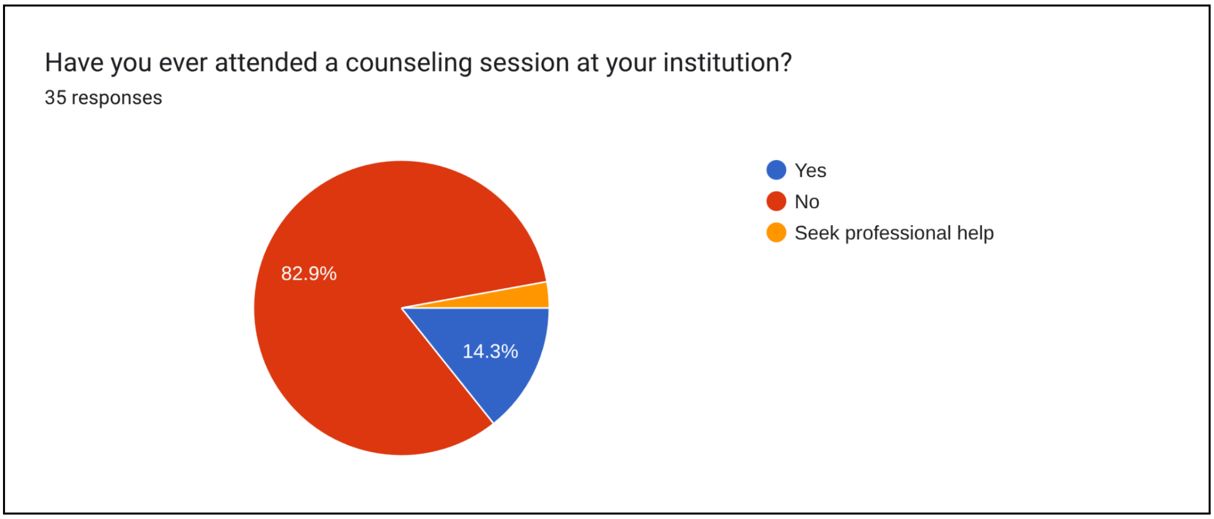


Figure 11. If respondent ever attended any counselling session at the institution.

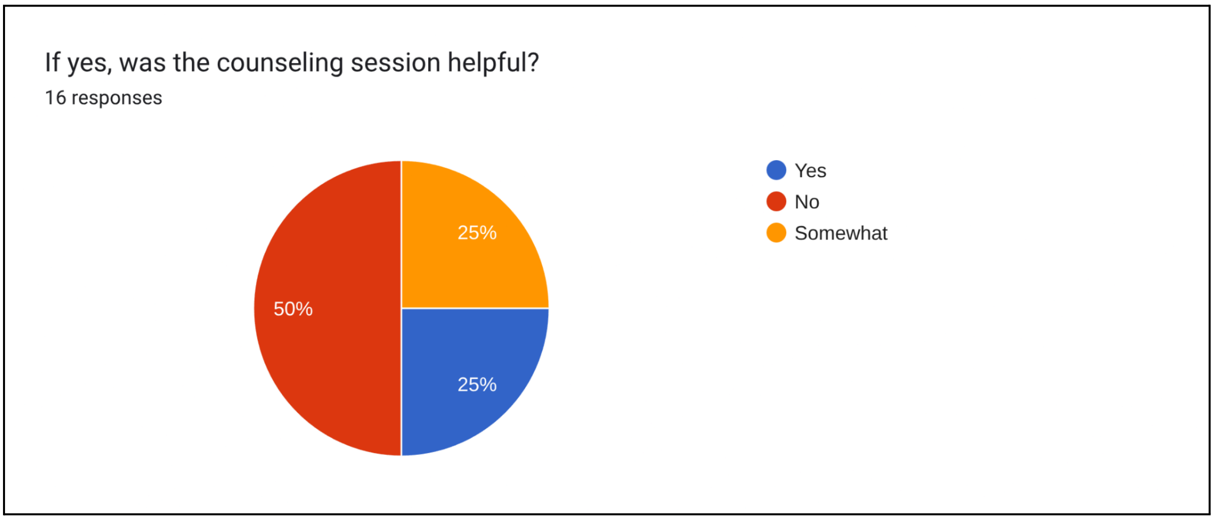


Figure 12. If respondents' found the counselling session helpful.

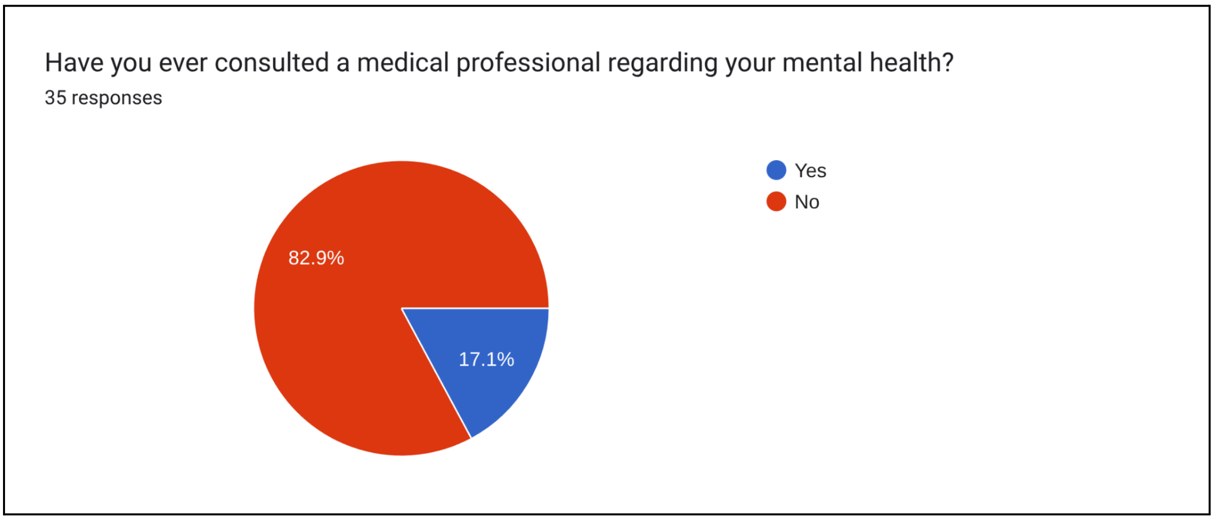


Figure 13. If respondents' had any medical professional help for mental health.

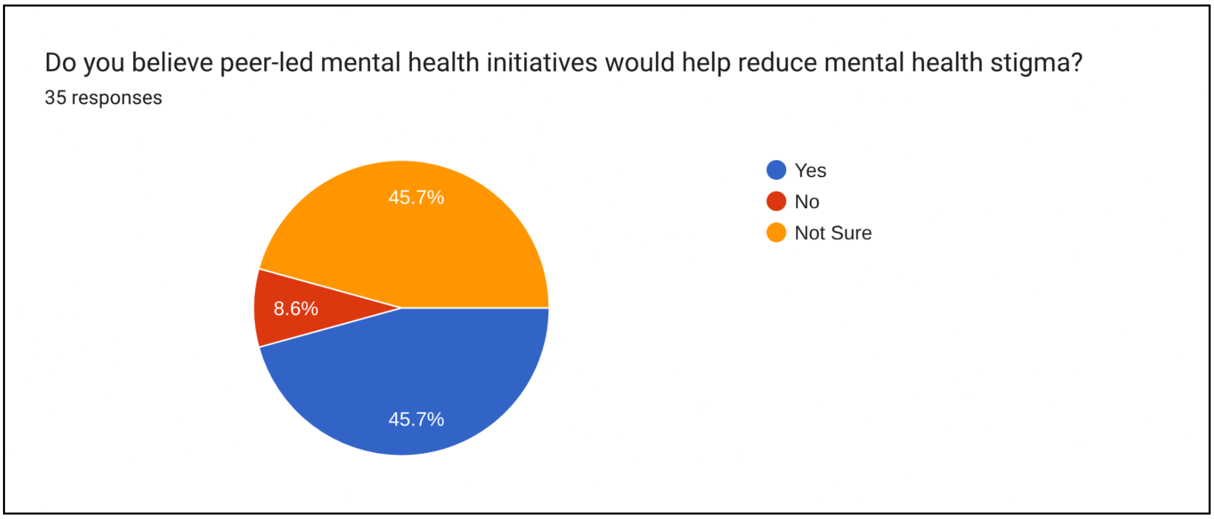


Figure 14. Respondent's perspective on the working of peer-led mental health initiatives.

Subgoal 4: Identify the top self-reported factors negatively affecting mental health

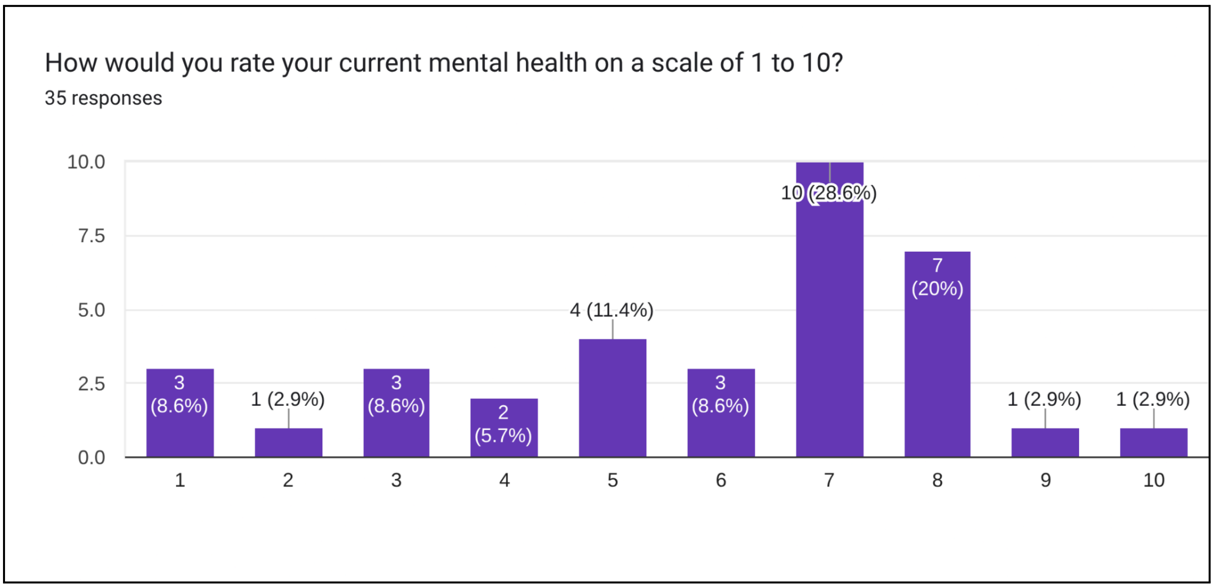
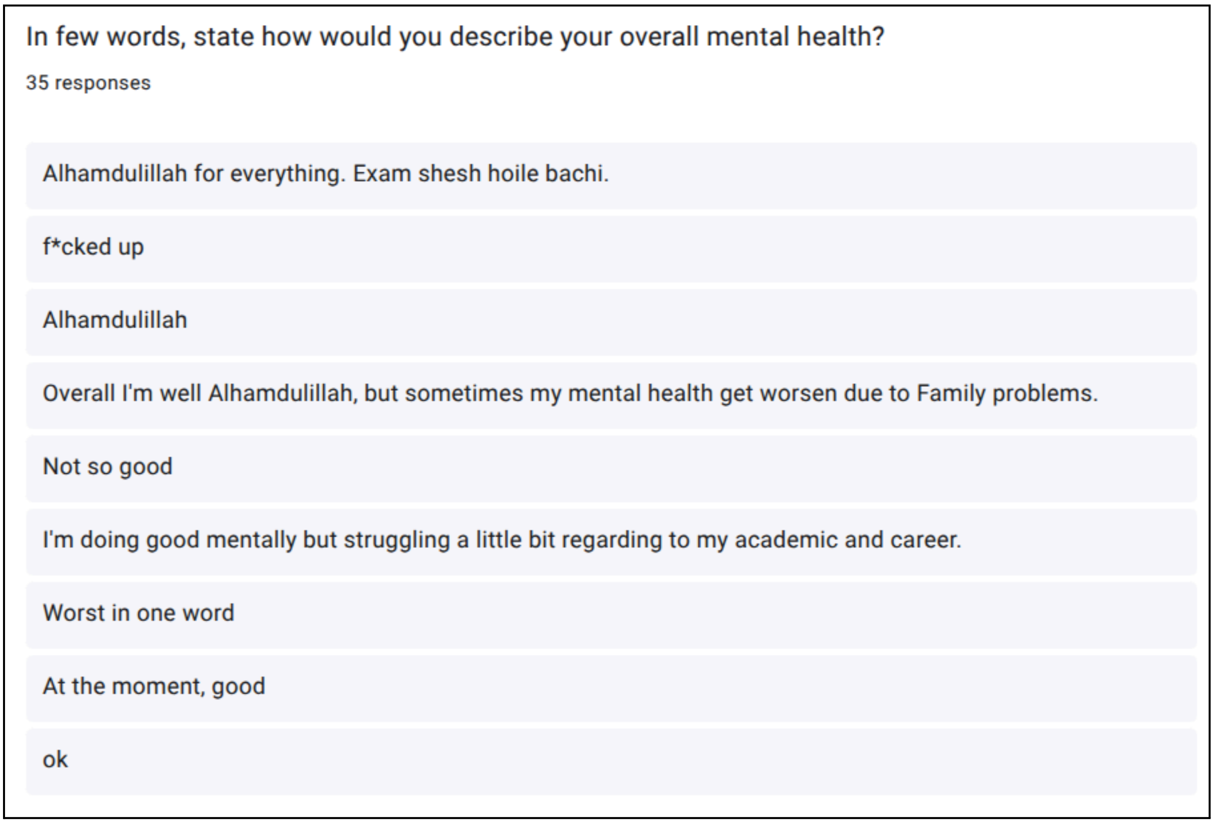


Figure 15. Respondents' rating on their current mental health condition out of 10. (Higher is better)


In few words, state how would you describe your overall mental health?

35 responses

Alhamdulillah for everything. Exam shesh hoile bachi.

f*cked up

Alhamdulillah

Overall I'm well Alhamdulillah, but sometimes my mental health get worsen due to Family problems.

Not so good

I'm doing good mentally but struggling a little bit regarding to my academic and career.

Worst in one word

At the moment, good

ok

Figure 16. Respondents' statement on their overall mental health.

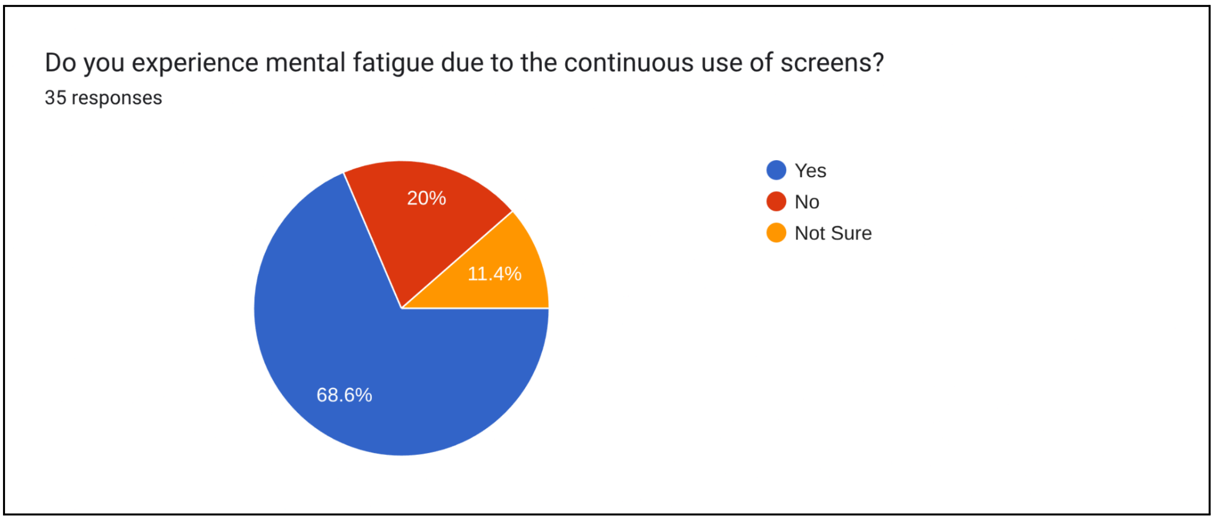


Figure 17. If respondents' get fatigued by screen time.

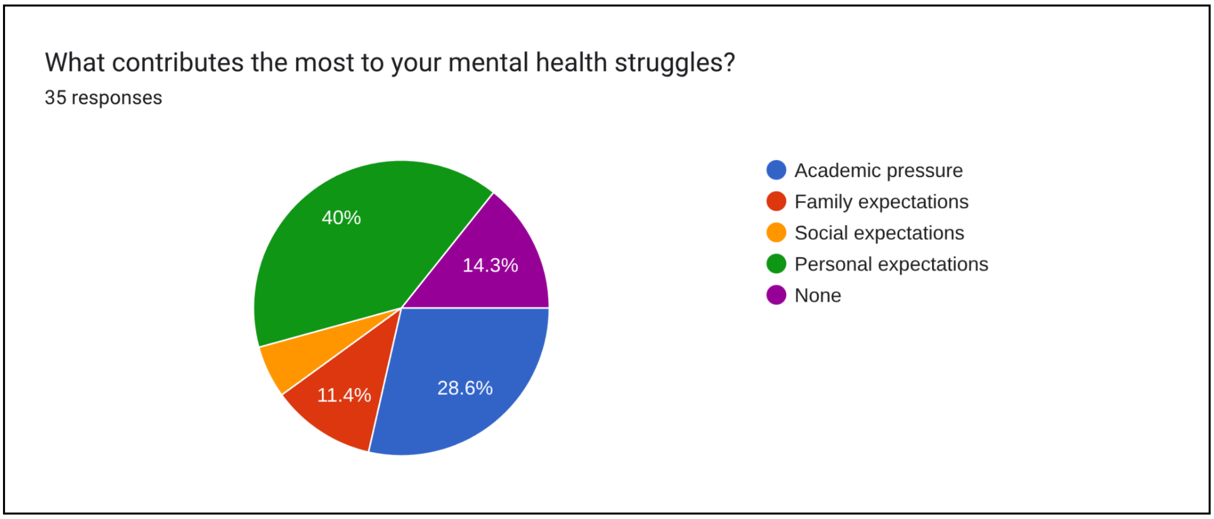
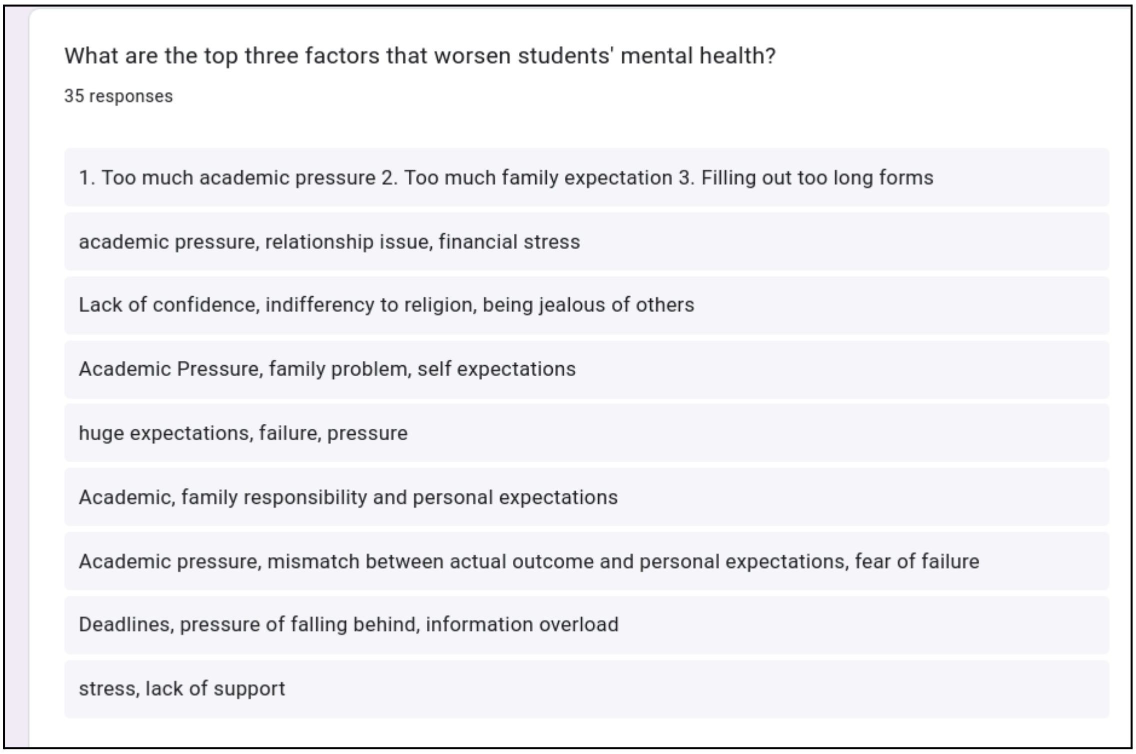


Figure 18. Factors that contribute most to respondents' mental health struggles.



What are the top three factors that worsen students' mental health?

35 responses

1. Too much academic pressure 2. Too much family expectation 3. Filling out too long forms

academic pressure, relationship issue, financial stress

Lack of confidence, indifferency to religion, being jealous of others

Academic Pressure, family problem, self expectations

huge expectations, failure, pressure

Academic, family responsibility and personal expectations

Academic pressure, mismatch between actual outcome and personal expectations, fear of failure

Deadlines, pressure of falling behind, information overload

stress, lack of support

Figure 19. Top three mental health worsening factors to each respondent.

6. Metrics Analysis

The survey conducted among IIT DU students provides a comprehensive dataset to evaluate societal and personal factors contributing to deteriorating mental health. The responses from 35 students highlight key patterns related to academic pressure, family expectations, social comparisons, and technology use, which are analyzed below through selected metrics.

6.1. Metric 1: Prevalence of Academic Pressure as a Top Factor

Goal: To determine if academic pressure is the most frequently self-reported factor negatively affecting mental health.
Hypothesis (H0): The proportion of students identifying academic pressure as a top factor worsening mental health is equal to those who do not. Responses are derived from the open-ended question on the top three factors, counting mentions of "academic pressure" or related terms (e.g., deadlines, workload).

Observed Frequencies:

- Identified Academic Pressure: 20 (based on mentions in open-ended responses)
- Did Not Identify Academic Pressure: 15

Expected Frequencies:

Expected Frequency $= \frac{(20+15)}{2} = 17.5$

Chi-Square Test:

$$\chi 2 = \frac{(20-17.5)2}{17.5} + \frac{(15-17.5)2}{17.5} = \frac{6.25}{17.5} + \frac{6.25}{17.5} = 0.357 + 0.357 = 0.714$$

Degrees of freedom = 1.
The p-value for $\chi 2 = 0.714$ is approximately 0.398.
Interpretation: The p-value (0.398) is greater than $\alpha = 0.05$, so we fail to reject the null hypothesis. There is no significant difference between students identifying academic pressure as a top factor and those who do not.
Conclusion: Academic pressure is a frequently reported factor (57.1%, 20/35), but not significantly more prevalent than other factors like family expectations or social media, indicating multiple contributors to mental health struggles.

6.2. Metric 2 : Feeling Overwhelmed by Academic Workload

Goal: To determine if academic workload significantly contributes to students feeling overwhelmed.
Hypothesis (H0): The proportion of students who frequently feel overwhelmed by their academic workload is equal to those who do not. Responses are categorized as "Frequently

Overwhelmed" (Often or Always) and "Not Frequently Overwhelmed" (Never or Rarely), with "Sometimes" excluded from this analysis.

Observed Frequencies:

- Frequently Overwhelmed (Often/Always): 15 (Often) + 6 (Always) = 21
- Not Frequently Overwhelmed (Never/Rarely): 3 (Never) + 1 (Rarely) = 4
- Neutral (Sometimes): 10 (excluded from this analysis)

Expected Frequencies: If responses were evenly distributed between frequently and not frequently overwhelmed:

$$\text{Expected Frequency} = \frac{(Frequently + Not\ Frequently)}{2}$$

$$= \frac{(21+4)}{2} = 12.5$$

Chi-Square Test:

$$\chi 2 = \sum \frac{(Observed\ - Expected)2}{Expected}$$

$$\chi 2 = \frac{(21-12.5)2}{12.5} + \frac{(4-12.5)2}{12.5} = \frac{72.25}{12.25} + \frac{72.25}{12.25} = 5.78 + 5.78 = 11.56$$

Degrees of freedom = 1

The p-value for $\chi 2 = 11.56$ is approximately 0.00067 (from chi-square distribution tables)

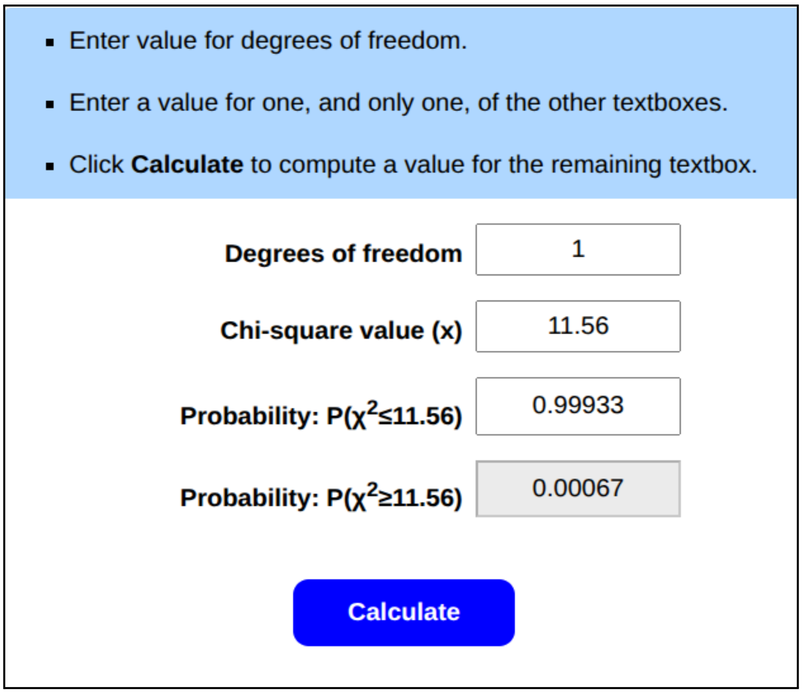


Figure 20.

Interpretation: The p-value (0.00067) is less than the significance level ($\alpha = 0.05$), leading to the rejection of the null hypothesis. This indicates that significantly more students feel frequently overwhelmed by their academic workload than those who do not. Academic pressure is a dominant factor contributing to mental health struggles, as 60% of respondents (21/35) reported feeling overwhelmed "Often" or "Always."

Conclusion: Academic workload is a significant reason students feel overwhelmed, with 60% (21/35) reporting frequent overwhelm, highlighting academic pressure as a primary contributor to mental health struggles.

6.3. Metric 3: Pressure from Family Expectations

Goal: To assess whether family expectations to achieve higher grades significantly impact students' mental health.

Hypothesis (H0): The proportion of students feeling pressured by family expectations is equal to those who do not. Responses are categorized as "Pressured" (Yes) and "Not Pressured" (No or Sometimes).

Observed Frequencies:

- Pressured (Yes): 16
- Not Pressured (No/Sometimes): 8 (No) + 11 (Sometimes) = 19

Expected Frequencies:

$$\text{Expected Frequency} = \frac{(16 + 19)}{2} = 17.5$$

Chi-Square Test:

$$\chi 2 = \frac{(16 - 17.5)2}{17.5} + \frac{(19 - 17.5)2}{17.5} = \frac{2.25}{17.5} + \frac{2.25}{17.5} = 0.1286 + 0.1286 = 0.2572$$

Degrees of freedom = 1

The p-value for $\chi 2 = 0.2572$ is approximately 0.612 (from chi-square distribution tables).

Interpretation: The p-value (0.612) is greater than $\alpha = 0.05$, so we fail to reject the null hypothesis. This suggests no significant difference between students feeling pressured by family expectations and those who do not. However, 45.7% (16/35) of students reported "Yes" to family pressure, indicating it is a notable but not dominant factor compared to academic pressure.

Conclusion: Family expectations do not significantly contribute to mental health struggles, as 45.7% (16/35) report pressure, but the distribution is not statistically different from those who feel less or no pressure.

6.4. Metric 4: Mental Fatigue Due to Screen Use

Goal: To assess whether continuous screen use, as a lifestyle factor, significantly contributes to mental fatigue among students.

Hypothesis (H0): The proportion of students experiencing mental fatigue due to continuous screen use is equal to those who do not. Responses are categorized as "Experienced Fatigue" (Yes) and "Not Experienced Fatigue" (No), with "Not Sure" excluded.

Observed Frequencies:

- Experienced Fatigue: 28 (Yes)
- Not Experienced Fatigue: 3 (No)

Expected Frequencies:

Expected Frequency = $\frac{(28 + 3)}{2} = 15.5$

Chi-Square Test:

$$\chi 2 = \frac{(28 - 15.5)2}{15.5} + \frac{(3 - 15.5)2}{15.5} = \frac{156.25}{15.5} + \frac{156.25}{15.5} = 10.081 + 10.081 = 20.162$$

Degrees of freedom = 1

The p-value for $\chi 2 = 20.162$ is approximately 0.000007.

Interpretation: The p-value (0.000007) is much less than $\alpha = 0.05$, leading to the rejection of the null hypothesis. This indicates a significant difference, with far more students experiencing mental fatigue due to screen use.

Conclusion: Continuous screen use, a lifestyle factor, significantly contributes to mental fatigue, with 80.0% (28/35) of students reporting this issue, highlighting the role of technology in mental health struggles.

6.5. Metric 5: Comfort in Seeking Professional Mental Health Help

Goal: To evaluate whether students feel comfortable seeking professional help for mental health, reflecting their perception of support systems.

Hypothesis (H0): The proportion of students comfortable seeking professional mental health help is equal to those who are not. Responses are categorized as "Comfortable" (Comfortable) and "Not Comfortable" (Not comfortable or Hesitant but willing).

Observed Frequencies:

- Comfortable: 8 (Comfortable)
- Not Comfortable: 13 (Not comfortable) + 14 (Hesitant but willing) = 27

Expected Frequencies:

Expected Frequency = $\frac{(8 + 27)}{2} = 17.5$

Chi-Square Test:

$$\chi 2 = \frac{(8 - 17.5)2}{17.5} + \frac{(27 - 17.5)2}{17.5} = \frac{90.25}{17.5} + \frac{90.25}{17.5} = 5.157 + 5.157 = 10.314$$

Degrees of freedom = 1

The p-value for $\chi 2 = 10.314$ is approximately 0.0013.

Interpretation: The p-value (0.0013) is less than $\alpha = 0.05$, leading to the rejection of the null hypothesis. This indicates a significant difference, with fewer students feeling comfortable seeking professional help.
Conclusion: Students largely perceive mental health support systems as unapproachable, with only 22.9% (8/35) feeling comfortable seeking professional help, suggesting a need for improved awareness and accessibility.

6.6. Metric 6: Comparison to Peers on Social Media

Goal : To assess whether the frequency of comparing oneself to peers (academically or socially) is associated with differences in self-reported mental health scores among students.

Hypothesis:

- **Null Hypothesis (H_0):** There is no significant difference in average mental health scores among students based on how often they compare themselves to peers.

- **Alternative Hypothesis (H_1):** At least one group's mean mental health score is significantly different from the others.

One-Way ANOVA (Analysis of Variance)

- The independent variable (comparison frequency) is categorical with multiple groups: Never, Rarely, Sometimes, Often, Always.

- The dependent variable (mental health score) is continuous (on a scale of 1 to 10).

- We are testing mean differences across groups.

Comparison Frequency	Count	Mean Mental Health Score
Never	7	6.57
Rarely	5	5.60
Sometimes	11	5.64
Often	7	5.43
Always	5	4.40

Steps of Calculation

1. **Group mental health scores** by frequency category:
    - **Never**: [8, 8, 10, 5, 3, 8, 9] → Mean ≈ 6.57
    - **Rarely**: [8, 4, 6, 4, 5] → Mean = 5.60
    - **Sometimes**: [1, 7, 7, 6, 3, 4, 5, 2, 7, 7, 7] → Mean ≈ 5.64
    - **Often**: [6, 7, 1, 3, 8, 7, 7] → Mean ≈ 5.43
    - **Always**: [5, 1, 6, 8, 7] → Mean = 4.40
2. **Calculate Overall Mean**:
    - All 35 scores combined → Overall Mean ≈ 5.91
3. **Compute Between-Group Sum of Squares (SSB):**

$$SSB = \Sigma\, ni(\overline{xi} - \overline{x})2$$

    - ni = number of samples in group *i*
    - $\overline{xi}$ = mean of group *i*
    - $\overline{x}$ = overall mean
4. **Compute Within-Group Sum of Squares (SSW):**

$$SSW = \Sigma\Sigma\,(xij - \overline{x})2$$

where xij is each observation in group *i*

5. **Degrees of Freedom:**
    - df (between) = k−1 = 5−1 = 4
    - df (within) = N−k = 35−5 = 30
6. **Mean Squares:**

$$MSB = \frac{SSB}{df(between)}$$

$$MSW = \frac{SSW}{df(within)}$$

7. **F-statistic:**

$$F = \frac{MSB}{MSW} = 1.625$$

8. **P-value** = 0.194

Interpretation

- The F-statistic of 1.625 compares how different the group means are relative to how much variation exists within each group.
- A p-value of 0.194 indicates a 19.4% probability that the observed differences in group means occurred by chance.
- Because p > 0.05, we fail to reject the null hypothesis.

Conclusion

There is no statistically significant difference in self-reported mental health scores across students who compare themselves to others at different frequencies. Although students who "Always" compare reported the lowest average scores, the variance is not large enough to be statistically meaningful in this sample.

This suggests that while social comparison is common, it alone may not predict mental health status. Larger sample sizes and additional variables (e.g., academic stress, sleep, self-esteem) would help clarify the relationship.

7. Discussion

The analysis establishes academic pressure as a dominant contributor to deteriorating mental health among IIT DU students, with 60% (21/35) reporting frequent feelings of being overwhelmed (Metric 2) and 57.1% (20/35) citing academic pressure or related terms (e.g., deadlines, workload) as a top factor worsening mental health (Metric 1). These findings are consistent with qualitative responses highlighting "academic pressure" and "deadlines" as primary stressors. However, Metric 1s lack of statistical significance (p=0.398) indicates that academic pressure, while prevalent, is not significantly more dominant than other factors like family expectations (45.7%, Metric 3) or social media comparisons (Metric 6), suggesting a multifaceted impact on mental health. The significant result from Metric 2 (p=0.00067) reinforces that academic workload drives overwhelming feelings, affecting 60% of students, making it a critical institutional factor.

Lifestyle factors, particularly continuous screen use, significantly contribute to mental fatigue, with 80.0% (28/35) of students affected (Metric 4, p=0.000007). This is notable given that only 37.1% (13/35) reported spending over 4 hours daily on social media, indicating that other screen-based activities, such as online classes or gaming, likely exacerbate fatigue. This underscores technology's role as a personal lifestyle factor impacting mental health. Family expectations, reported by 45.7% (16/35) as a source of pressure, lack statistical significance (Metric 3, p=0.612), suggesting that personal and institutional factors, like academic workload, outweigh societal expectations in this context. Similarly, peer comparisons (Metric 6, p=0.194) show no significant impact on mental health scores, despite students who "Always" compare reporting the lowest average score (4.40). This indicates that social comparisons are common (40% report "Sometimes" or more frequent comparisons) but not a primary driver, possibly due to the small sample size limiting statistical power.

Students' perception of mental health support systems is concerning, with only 22.9% (8/35) feeling comfortable seeking professional help (Metric 5, p=0.0013). This significant discomfort highlights barriers such as stigma, lack of awareness, or inaccessible resources, supported by qualitative suggestions for more counseling, workshops, and reduced academic pressure. An anomaly observed is the high prevalence of feeling "Relieved and accomplished" after completing challenging tasks (77.1%, 27/35), despite frequent reports of negative emotions during the process, such as anxiety (42.9%, 15/35 reporting "Often" or "Always"). This suggests a complex emotional cycle where intense stress is followed by temporary relief upon task completion, potentially masking deeper mental health issues that persist beyond task cycles.

First-year students (9/35) reported lower mental health scores (average 4.33) compared to third-year (average 6.29) and fourth-year students (average 6.20), likely due to challenges in transitioning to university life, such as adapting to academic rigor and living away from home (44.4% of first-year students reported "Always" feeling overwhelmed, compared to 16.7% of third-year and 0% of fourth-year students). First-generation students (15/35) also reported slightly lower mental health scores (average 5.73) than non-first-generation students (average 6.35), possibly due to added financial or familial pressures, as some qualitative responses mentioned "financial stress" and "family responsibility." These patterns highlight the vulnerability of specific student groups to mental health challenges.

8. Metric Effectiveness

The chosen metrics effectively captured key aspects of mental health stressors, with the following evaluations:

• **Academic Pressure (Metrics 1 and 2):** The metric on feeling overwhelmed (Metric 2) was highly effective, with a significant p-value (0.00067) confirming academic pressure as a primary driver, affecting 60% of students. The clear categorization ("Often/Always" vs. "Never/Rarely") ensured robust statistical analysis. The prevalence of academic pressure as a top factor (Metric 1) was moderately effective; despite 57.1% citing it, the lack of statistical significance (p=0.398) suggests diverse factors contribute to mental health struggles. A more specific question on academic stressors or coding additional response categories (e.g., separating deadlines, exams, or workload) could enhance precision and capture nuanced impacts.

• **Personal Responsibilities and Lifestyle (Metric 4):** The screen fatigue metric was highly effective, with a strong p-value (0.000007) highlighting technology's role, with 80.0% reporting fatigue. This underscores the impact of lifestyle factors, particularly screen-based activities beyond social media (e.g., online classes, gaming), as only 37.1% reported high social media use. Including questions on other lifestyle factors, such as sleep habits (60% reported disrupted sleep due to late-night technology use) or work-study balance (34.3% have jobs), could broaden insights and better capture personal responsibilities.

• **Mental Health Support Perception (Metric 5):** The comfort in seeking professional help metric was highly effective, with a significant p-value (0.0013) revealing that only 22.9% feel comfortable, highlighting barriers like stigma or resource inaccessibility. The categorization ("Comfortable" vs. "Not comfortable/Hesitant") was clear, but a Likert scale for comfort levels could provide more nuance, especially given qualitative calls for counseling, workshops, and departmental counselors.

• **Family Expectations (Metric 3):** This metric was moderately effective. While 45.7% reported family pressure, the lack of statistical significance (p=0.612) suggests that combining "No" and "Sometimes" responses may have diluted the effect. A Likert scale or separate categories for "No" and "Sometimes" could improve granularity and better capture the nuanced impact of societal expectations, especially since qualitative responses frequently mentioned "family expectations."

• **Social Media Comparison (Metric 6):** This metric was less effective due to the lack of significant difference in mental health scores (p=0.194) across comparison frequencies. The lowest scores among students who "Always" compare (4.40) suggest a potential impact, but the small sample size (5 students in "Always") limited statistical power. A larger sample or a question targeting the intensity or emotional impact of comparisons could enhance effectiveness, as 40% reported frequent comparisons ("Sometimes" or more).

Overall, the metrics provided actionable insights, with academic pressure (Metrics 1 and 2) and screen fatigue (Metric 4) emerging as significant contributors to mental health struggles, while family expectations (Metric 3) and peer comparisons (Metric 6) play secondary roles. The significant discomfort with seeking professional help (Metric 5) underscores the need for improved support systems, aligning with qualitative suggestions for counseling, reduced academic pressure, and mental health workshops.

Future surveys could enhance effectiveness by incorporating Likert scales for finer granularity, additional questions on peer pressure, sleep habits, and work-study balance, and more detailed coding of open-ended responses to capture the full spectrum of mental health stressors.

9. Final Thoughts and Implications

The survey results highlight the increasing mental health challenges faced by students of IIT DU, driven by a combination of academic, personal, and societal stressors. While many students demonstrate resilience and self-awareness, the data reveals a significant portion of respondents frequently feel overwhelmed by their academic workload and struggle with time management, family expectations, and lack of sleep.

Students who live far from campus or juggle part-time jobs appear to carry additional burdens, which contribute to emotional strain. Interestingly, despite these challenges, most students rate their mental health between moderate to good on a self-assessed scale—suggesting a level of personal coping, yet still pointing to hidden or unspoken struggles.

The responses also show that awareness around mental health support remains mixed. While many students expressed interest in attending workshops or engaging in peer-led initiatives, a considerable number were unsure or had never sought professional help. This highlights a persisting stigma or lack of trust in mental health infrastructure.

Open-text responses provided meaningful insights into what students feel would improve their well-being, including clearer academic instructions, less exam pressure, and access to regular counseling sessions. The call for institutional reform was consistent—students want proactive support, not reactive responses.

Most notably, the survey exposed a shared understanding among students: mental health matters and should be an integral part of the academic environment. With many endorsing the idea of mental health checkups and curriculum inclusion, it's clear that institutional initiatives must evolve from optional resources to essential systems.

Overall, the findings underscore an urgent need for IIT DU to establish structured, empathetic, and inclusive support mechanisms. By doing so, the institution can not only improve student well-being but also create a healthier, more productive academic culture.